EVALUATION IN MEDICAL IMAGE PROCESSING

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BINARY CLASS CLASSIFICATION

		True Class (From GT)	
		Class 1	Class 2
Predicted class (From algorithm)	Class 1	True Positive (TP)	False Positive (FP)
	Class 2	False Negative (FN)	True Negative (TN)

MULTICLASS CLASSIFICATION

	True Class (From GT)				
Predicted class (From algorithm)		Class 1	Class 2	Class 3	Class 4
	Class 1	TP1	a	b	c
	Class 2	d	TP2	e	f
	Class 3	g	h	TP3	i
	Class 4	j	k	l	TP4

Class 1	Actual Values	
Predicte d Values	TP1	FP
	FN	TN

MULTICLASS CLASSIFICATION

	True Class (From GT)				
		Class 1	Class 2	Class 3	Class 4
Predicted class (From algorithm)	Class 1	TP1	a	b	c
	Class 2	d	TP2	e	f
	Class 3	g	h	TP3	i
	Class 4	j	k	l	TP4

Class 2	Actual Values	
Predicte d Values	TP1	FP
	FN	TN

METRICES

Predicted class (From algorithm) \ True Class (From GT)		
	True Positive (TP)	False Positive (FP)
	False Negative (FN)	True Negative (TN)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall\ (Sens) = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$Mean\ of\ Sens\ and\ Spec\ (MSS) = \frac{sens + spec}{2}$$

- Accuracy: how well a diagnostic test or model correctly identifies both the TP and TN
- Precision: the ability of a diagnostic test or model to correctly identify positive cases while minimizing false positives.
- Recall: the ability of a diagnostic test or model to correctly identify positive cases, including true positives
- Specificity: the ability of a diagnostic test or model to correctly identify negative cases.
- F1-Score: balance between precision and recall.

METRICES

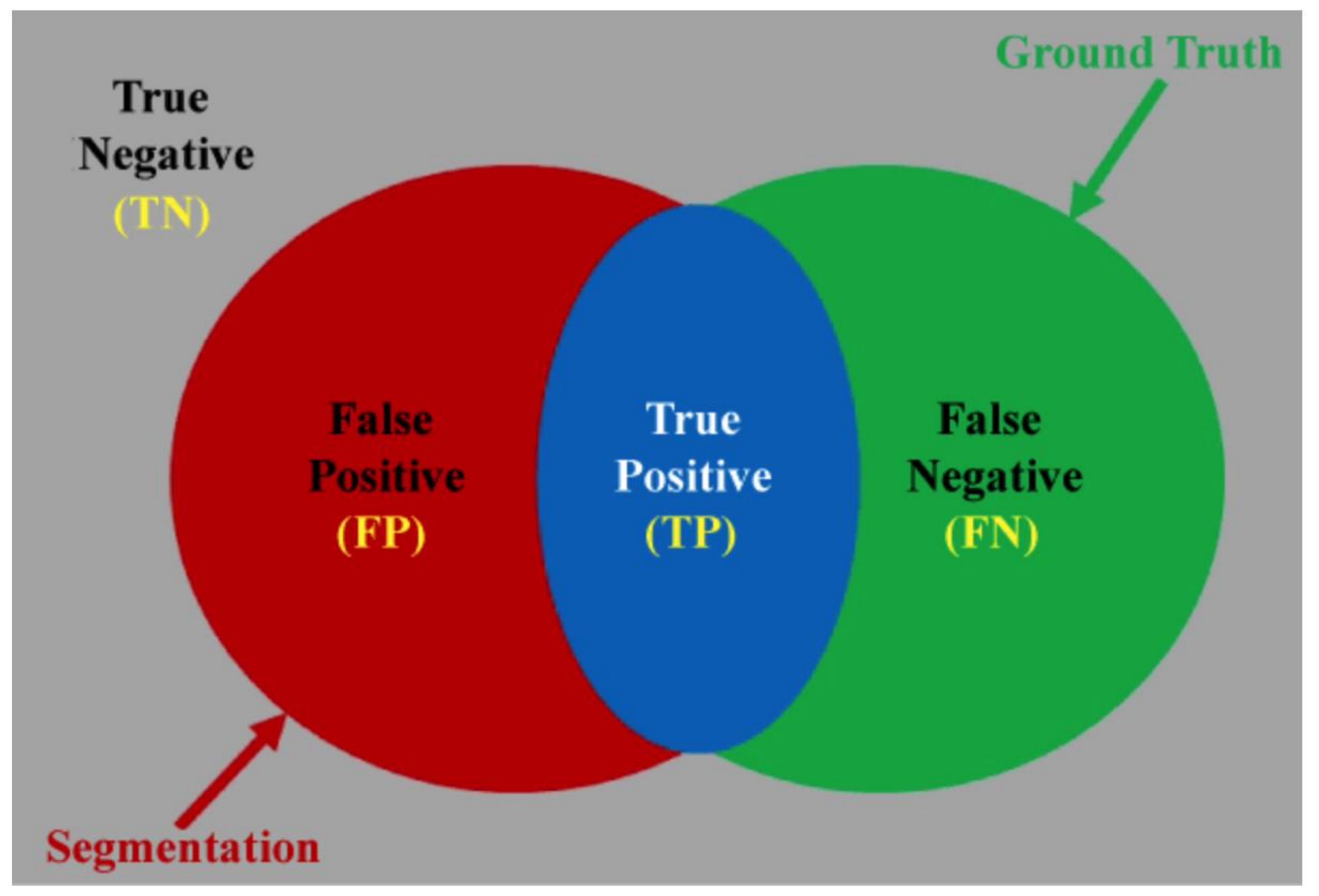
- In medical diagnosis, two metrices are important:
  - High Sensitivity → Cannot miss patient with disease (low FN)
  - High Specificity → Cannot misdiagnose healthy patients as a patient with disease (low FP)
- Most medical dataset are imbalanced → models tend to overfitting → the accuracy “might be” good

$$Mean\ of\ Sens\ and\ Spec\ (MSS) = \frac{sens + spec}{2}$$

WHEN WE USE MSS?

Use MSS if …	Consider other metrics if …
Binary classification	Complex segmentation or multiclass problems
Imbalance data	Need a metric focused on one side
Want a simple and balanced evaluation	Want detailed analysis per case or threshold

REGION BASED METRICS



		True Class (From GT)	
Predicted class (From algorithm)		True Positive (TP)	False Positive (FP)
		False Negative (FN)	True Negative (TN)

$$TP = \frac{\left|O_{SR}^{M} \cap O_{GT}\right|}{\left|O_{GT}\right|}$$

$$TN = \frac{\left|\left(O_{SR}^{M} \cup O_{GT}\right)^{c}\right|}{\left|\left(O_{GT}\right)^{c}\right|}$$

- O_{SR}^{M}: Segmentation result
- O_{GT}: Ground Truth

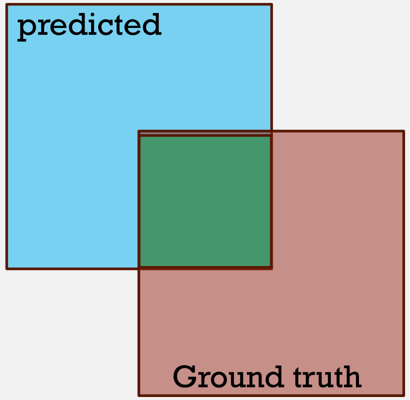
REGION BASED METRICES

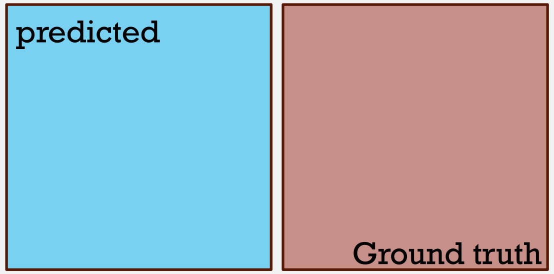
$$Dice\ Similarity = \frac{2\left|O_{GT} \cap O_{SR}^{M}\right|}{\left|O_{GT}\right| + \left|O_{SR}^{M}\right|}$$

$$Jaccard\ Index\ (IoU) = \frac{\left|O_{GT} \cap O_{SR}^{M}\right|}{\left|O_{GT} \cup O_{SR}^{M}\right|}$$

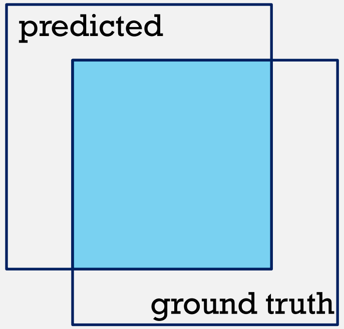
Dice Coefficient = $\dfrac{\text{2x area of overlapped}}{\text{Total area}}$ =

2x





IoU = $\dfrac{\text{Area of Overlap}}{\text{Area of Union}}$



THANK YOU